

ER DIAGRAM & RELATIONAL SCHEMA DESIGN

HOSPITAL MANAGEMENT SYSTEM

Modelling the hospital management application as a complete Entity-Relationship diagram and converting it into a normalised relational schema.

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Course	BCA AI/ML
Subject	Database Management System
Assignment	Assignment 2 — ER Diagram & Relational Schema

01 TASK 1 Entities, Attributes & Primary Keys

The core entities of the Hospital Management System, carried forward from the first DBMS assignment, with their attributes and primary keys identified.

◆ PATIENT

Attribute	Data Type	Key	Description
patient_id	INT	PK	Unique identifier for each patient
name	VARCHAR(100)	—	Full name of the patient
gender	VARCHAR(10)	—	Gender of the patient
dob	DATE	—	Date of birth, used to compute age
phone	VARCHAR(15)	—	Contact phone number
address	VARCHAR(255)	—	Residential address

◆ DEPARTMENT

Attribute	Data Type	Key	Description
department_id	INT	PK	Unique identifier for each department
department_name	VARCHAR(80)	—	Name of the department, e.g. Cardiology
location	VARCHAR(100)	—	Block / floor where the department is situated

◆ DOCTOR

Attribute	Data Type	Key	Description
doctor_id	INT	PK	Unique identifier for each doctor
name	VARCHAR(100)	—	Full name of the doctor
specialization	VARCHAR(80)	—	Medical specialization, e.g. Cardiologist
phone	VARCHAR(15)	—	Contact phone number
email	VARCHAR(120)	—	Contact email

◆ APPOINTMENT

Attribute	Data Type	Key	Description
appointment_id	INT	PK	Unique identifier for each appointment
appointment_date	DATE	—	Date the appointment is scheduled for
appointment_time	TIME	—	Time the appointment is scheduled for
status	VARCHAR(30)	—	Scheduled / Completed / Cancelled / No-show

◆ MEDICINE

Attribute	Data Type	Key	Description
medicine_id	INT	PK	Unique identifier for each medicine
medicine_name	VARCHAR(120)	—	Name of the medicine
manufacturer	VARCHAR(100)	—	Manufacturer / pharmaceutical company
price	DECIMAL(10,2)	—	Unit price of the medicine

◆ BILL

Attribute	Data Type	Key	Description
bill_id	INT	PK	Unique identifier for each bill
bill_date	DATETIME	—	Date and time the bill was generated
total_amount	DECIMAL(10,2)	—	Total payable amount
payment_status	VARCHAR(30)	—	Pending / Paid / Partially Paid / Refunded

◆ FEEDBACK

Attribute	Data Type	Key	Description
feedback_id	INT	PK	Unique identifier for each feedback entry
rating	INT	—	Rating given by the patient, 1 to 5
comments	VARCHAR(500)	—	Written feedback from the patient

Note: linking columns such as patient_id in Appointment or appointment_id in Bill are deliberately not listed as entity attributes here. At the ER level they are expressed as relationships (Section 2) and only appear as foreign keys once the schema is derived in Section 4.

02 TASK 2 Relationships & Cardinalities

Seven relationships connect the seven entities above. Six are one-to-many; one — Appointment and Medicine — is many-to-many and is resolved into its own table in Section 4.

Relationship	Participating Entities	Cardinality	Meaning
REGISTERS	Patient → Appointment	1 : M	A patient may book many appointments; each appointment is booked by exactly one patient.
WORKS IN	Department → Doctor	1 : M	A department can employ many doctors; each doctor belongs to exactly one department.
ATTENDS	Doctor → Appointment	1 : M	A doctor can attend many appointments over time; each appointment is attended by exactly one doctor.
PRESCRIBES	Appointment ↔ Medicine	M : N	An appointment can result in several medicines being prescribed, and a medicine can be prescribed across many appointments. Carries its own attributes (dosage, duration, quantity).

Relationship	Participating Entities	Cardinality	Meaning
BILLED AS	Appointment → Bill	1 : M	An appointment can generate more than one bill record, e.g. consultation fee, lab charges or a later revision; each bill belongs to one appointment.
WRITES	Patient → Feedback	1 : M	A patient can write many feedback entries; each feedback entry is written by exactly one patient.
RECEIVES	Doctor → Feedback	1 : M	A doctor can receive many feedback entries from different patients; each feedback entry refers to exactly one doctor.

Note: no one-to-one (1:1) relationship occurs naturally in this domain — every association here either links one record to many related records, or (in the case of PRESCRIBES) links many records on each side.

03 TASK 3 Complete ER Diagram

Drawn in Chen notation — entities as rectangles, relationships as diamonds, attributes as ovals, and primary keys underlined.

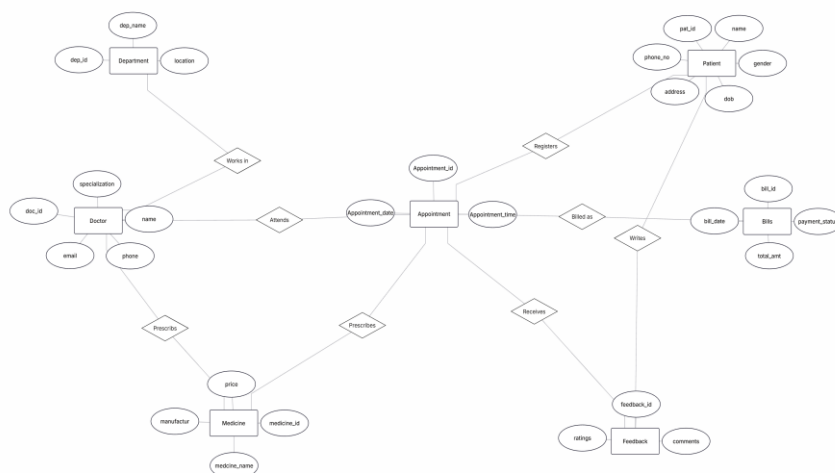


Figure 1 — Entity-Relationship Diagram of the Hospital Management System (Chen notation).

04 TASK 4 Relational Schema

The ER diagram is converted into tables using three standard mapping rules.

- **Entities become tables:** each entity turns into a table, and its attributes become columns; the entity's primary key stays the table's primary key.
- **1:M relationships become foreign keys:** the primary key of the "one" side is placed as a foreign key on the table of the "many" side (e.g. department_id on Doctors, patient_id and doctor_id on Appointments).

- **The M:N relationship becomes a new table:** PRESCRIBES is converted into an associative table, Prescriptions, holding the primary keys of both Appointments and Medicines as foreign keys, plus the relationship's own attributes (dosage, duration, quantity) and its own surrogate primary key, prescription_id.

Figure 2 — Relational Schema of the Hospital Management System, showing primary and foreign keys

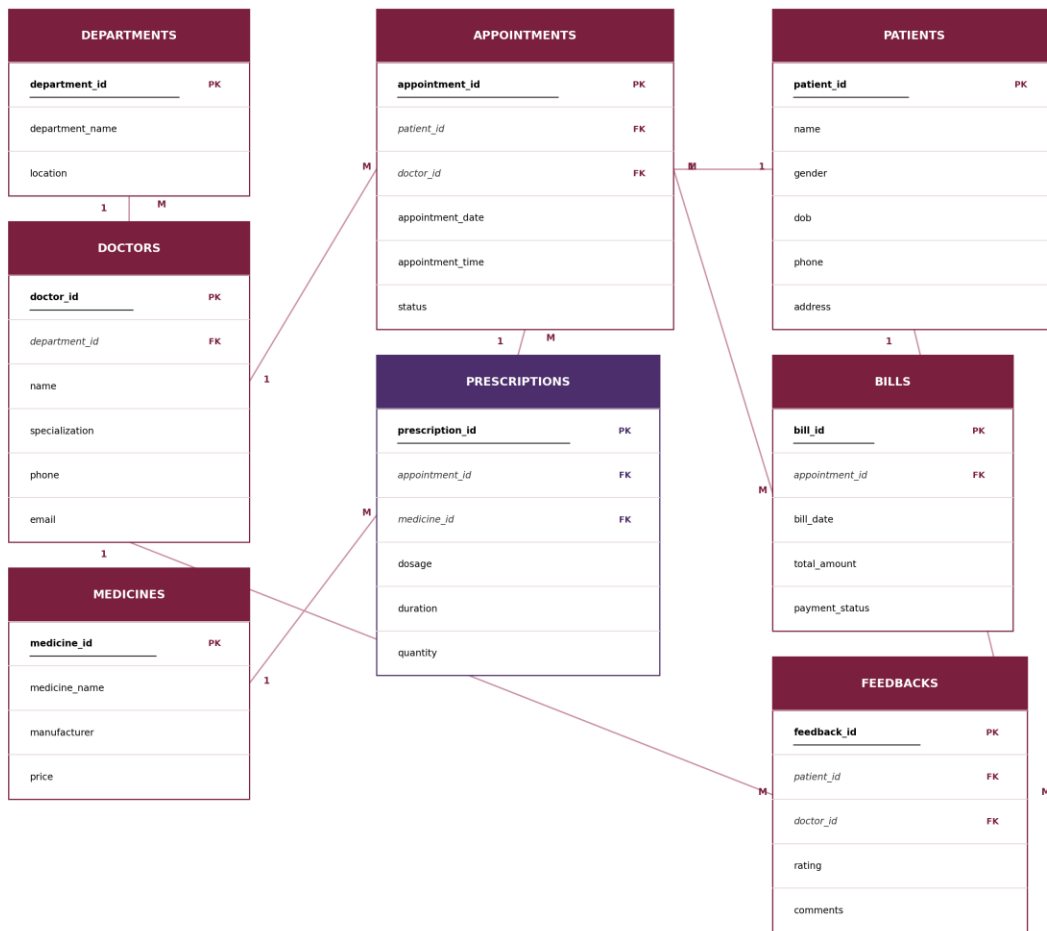


Figure 2 — Relational Schema of the Hospital Management System, showing primary and foreign keys.

◆ PATIENTS

Column	Type	Key	Description
patient_id	INT	PK	Unique patient identifier
name	VARCHAR(100)	—	Patient full name
gender	VARCHAR(10)	—	Gender of the patient
dob	DATE	—	Date of birth
phone	VARCHAR(15)	—	Contact number
address	VARCHAR(255)	—	Residential address

◆ DEPARTMENTS

Column	Type	Key	Description
department_id	INT	PK	Unique department identifier
department_name	VARCHAR(80)	—	Department name, e.g. Cardiology
location	VARCHAR(100)	—	Block / floor location

◆ DOCTORS

Column	Type	Key	Description
doctor_id	INT	PK	Unique doctor identifier
<i>department_id</i>	INT	FK	References Departments.department_id
name	VARCHAR(100)	—	Doctor full name
specialization	VARCHAR(80)	—	Medical specialization
phone	VARCHAR(15)	—	Contact number
email	VARCHAR(120)	—	Contact email

◆ APPOINTMENTS

Column	Type	Key	Description
appointment_id	INT	PK	Unique appointment identifier
<i>patient_id</i>	INT	FK	References Patients.patient_id
<i>doctor_id</i>	INT	FK	References Doctors.doctor_id
appointment_date	DATE	—	Date of the appointment
appointment_time	TIME	—	Time of the appointment
status	VARCHAR(30)	—	Scheduled / Completed / Cancelled / No-show

◆ PRESCRIPTIONS (RESOLVES THE M:N RELATIONSHIP)

Column	Type	Key	Description
prescription_id	INT	PK	Unique prescription-line identifier
<i>appointment_id</i>	INT	FK	References Appointments.appointment_id
<i>medicine_id</i>	INT	FK	References Medicines.medicine_id
dosage	VARCHAR(50)	—	Dosage prescribed, e.g. 500mg twice daily
duration	VARCHAR(50)	—	Duration of the course, e.g. 5 days
quantity	INT	—	Number of units prescribed

◆ MEDICINES

Column	Type	Key	Description
medicine_id	INT	PK	Unique medicine identifier
medicine_name	VARCHAR(120)	—	Name of the medicine
manufacturer	VARCHAR(100)	—	Manufacturer / pharmaceutical company
price	DECIMAL(10,2)	—	Unit price

◆ BILLS

Column	Type	Key	Description
bill_id	INT	PK	Unique bill identifier
<i>appointment_id</i>	INT	FK	References Appointments.appointment_id
bill_date	DATETIME	—	Date and time of billing
total_amount	DECIMAL(10,2)	—	Total payable amount
payment_status	VARCHAR(30)	—	Pending / Paid / Partially Paid / Refunded

◆ FEEDBACKS

Column	Type	Key	Description
feedback_id	INT	PK	Unique feedback identifier
<i>patient_id</i>	INT	FK	References Patients.patient_id
<i>doctor_id</i>	INT	FK	References Doctors.doctor_id
rating	INT	—	Rating, 1 to 5
comments	VARCHAR(500)	—	Written feedback

05 TASK 5 Relationship Explanation

A plain-language walk-through of each relationship in the ER diagram and why it carries the cardinality it does.

- **Patient registers Appointment (1:M)** — each appointment is tied to exactly one patient, but a patient's appointment history can grow without limit, so the "many" side sits on Appointment.

- ▶ **Department employs Doctor (1:M)** — a department such as Cardiology groups many doctors, while every doctor is attached to a single department to keep staffing unambiguous.
- ▶ **Doctor attends Appointment (1:M)** — a doctor's diary fills up with many appointments over time, but any single appointment is handled by exactly one doctor.
- ▶ **Appointment prescribes Medicine (M:N)** — an appointment can result in several medicines being prescribed together, and a common medicine such as paracetamol is naturally prescribed across many different appointments, so neither side can be capped at one. The dosage, duration and quantity of each line are properties of the pairing itself, not of the appointment or the medicine alone, which is why they sit on the relationship rather than on either entity.
- ▶ **Appointment billed as Bill (1:M)** — one bill usually settles an appointment, but the model allows more than one bill row per appointment to capture consultation fees, lab charges or refunds, without ever letting one bill settle two appointments.
- ▶ **Patient writes Feedback (1:M)** — a patient may leave feedback after several visits over time, but any one feedback entry is authored by a single patient.
- ▶ **Doctor receives Feedback (1:M)** — a popular doctor accumulates many feedback entries from different patients, while each entry naturally evaluates just one doctor.

06 SUMMARY Conclusion

The Hospital Management System is modelled as seven entities connected by seven relationships. Six relationships are one-to-many; the Appointment–Medicine relationship is many-to-many and, following standard ER-to-relational conversion rules, is resolved into its own associative table, Prescriptions. This mirrors exactly how the first assignment's schema was built, confirming that the conceptual ER design and the working relational schema describe the same eight-table database.

Every table carries a single-column surrogate primary key, and every one-to-many relationship is implemented as a foreign key on the "many" side, keeping referential integrity straightforward to enforce and the design easy to extend.

Expected Output Status

Expected Output	Status
List of entities and attributes	✓ Included
ER Diagram	✓ Included
Relationships and cardinalities	✓ Included
Relational Schema	✓ Included
Primary and Foreign Key identification	✓ Included
Brief relationship explanation	✓ Included